Query reformulation across increasing colloquiality

R7 held-out comparison · fixed retrieval budgets · descriptive point estimates

Held-out set: 8 question families, 16 bilingual queries per level, 6 document components from two textbooks.
AI-authored variants and AI-reviewed labels; local document/topic holdout, not unseen-textbook or real-user validation.
A–C exclude L4. D evaluates ambiguity separately. Point estimates only; paired component intervals are in the report.
C uses one shared reranking request over the candidate union; each policy retains at most 20 candidates and a 1,024-token budget.
Shared scoring is a controlled experiment, not a production request-budget or end-to-end latency measurement.